

# Project Overview

The goal of this internship is to transform an automatically generated AI4RSE taxonomy into a validated ontology that accurately represents the domain of Artificial Intelligence for Research Software Engineering (AI4RSE).

As part of previous research activities, we analyzed a large collection of scientific publications and extracted approximately 11,000 concepts related to Artificial Intelligence and Research Software Engineering. These concepts were automatically categorized using Large Language Models (LLMs) and aligned with the IEEE Taxonomy (2025 edition). The result is an initial taxonomy that organizes concepts into a hierarchical structure consisting of High-Level Categories, Middle-Level Categories, Low-Level Categories, and Concepts.

Although the taxonomy has already been generated, its quality has not yet been systematically evaluated. Since the taxonomy was created using automated techniques, some concepts may be assigned to inappropriate categories, duplicated across different locations, insufficiently defined, or only partially relevant to the AI4RSE domain. Consequently, the taxonomy requires validation and refinement before it can be used as the basis for a more sophisticated knowledge representation.

The internship consists of three interconnected tasks that collectively form a complete knowledge-engineering pipeline, starting with taxonomy validation and ending with ontology evaluation.

## Task 1: Evaluation and Refinement of the AI4RSE Taxonomy (Weeks 1–2)

The first task focuses on evaluating the quality of the existing taxonomy and refining the current concept-category alignment. The objective is to determine whether concepts have been assigned to appropriate categories and identify concepts that require modification.

We will use Generative AI as an assistant to evaluate concept-category alignment and identify potential issues such as misclassified concepts, duplicate concepts, ambiguous concepts, overly generic concepts, and concepts that are not sufficiently relevant to the AI4RSE domain.

All decisions will be recorded in a structured Google Sheet using a small set of predefined actions such as Keep, Move, Merge, Remove, and Discuss. The resulting spreadsheet will serve both as documentation of the evaluation process and as input for an automated taxonomy-refinement pipeline.

The outcome of this task will be a refined taxonomy with improved concept-category alignment and a structured set of recommendations describing how the taxonomy should be updated.

**Duration:** 2 weeks

**Expected Outputs:**

- Refined AI4RSE taxonomy
- Taxonomy review Google Sheet
- Summary of identified taxonomy issues
- Taxonomy refinement recommendations

## Task 2: Ontology Construction from the Refined Taxonomy (Weeks 3–5)

The second task focuses on transforming the refined taxonomy into an ontology.

While a taxonomy primarily organizes concepts into hierarchical categories, an ontology provides a richer representation of knowledge by explicitly defining concepts, categories, properties, and semantic relationships.

During this task, we will identify ontology classes and subclasses based on the refined taxonomy. We will then define semantic relationships between concepts and categories, establish provenance links to source publications, and construct a machine-readable representation of the AI4RSE domain.

Examples of semantic relationships may include:

- supports
- automates
- improves
- depends on
- uses
- generates
- evaluates

The ontology should provide a structured and extensible representation of AI4RSE knowledge that can support future research, analysis, and knowledge-discovery activities.

**Duration:** 3 weeks

**Expected Outputs:**

- AI4RSE ontology
- Ontology schema and documentation
- Relationship catalogue
- Ontology construction report

## Task 3: Ontology Evaluation and Validation (Weeks 6–8)

The third task focuses on evaluating the quality and usefulness of the ontology produced in Task 2.

The objective is to determine whether the ontology provides a meaningful and accurate representation of the AI4RSE domain.

The evaluation will include multiple perspectives, including structural validation, competency-question assessment, and expert review. Structural validation will focus on identifying issues such as duplicate concepts, orphan concepts, inconsistent relationships, and modeling errors. Competency questions will be used to determine whether the ontology can answer representative questions about the AI4RSE domain. Expert feedback will be collected to assess the correctness, completeness, and usefulness of the ontology.

The results of this evaluation will help identify strengths, weaknesses, and opportunities for future improvement.

**Duration:** 3 weeks

**Expected Outputs:**

- Ontology evaluation report
- Competency-question dataset
- Validation results
- Expert feedback summary
- Recommendations for future ontology improvements

## Final Deliverables

By the end of the internship, we expect to produce:

1. A refined AI4RSE taxonomy.
2. A structured taxonomy-review dataset (Google Sheet).
3. An AI4RSE ontology.
4. Ontology documentation.
5. Ontology evaluation results.
6. A final internship report describing the methodology, findings, and recommendations.

Overall, this internship follows a complete knowledge-engineering workflow:

**Automatically Generated Taxonomy → Refined Taxonomy → Ontology → Evaluated Ontology**

The resulting ontology will provide a structured representation of AI4RSE knowledge and can serve as a foundation for future research activities, knowledge graphs, semantic search systems, intelligent assistants, and decision-support tools for Research Software Engineering.

# Internship Task 1: Evaluation and Refinement of the AI4RSE Taxonomy

## 1. Introduction

Artificial Intelligence for Research Software Engineering (AI4RSE) is an emerging research area that investigates how Artificial Intelligence techniques can support the development, maintenance, testing, documentation, operation, and evolution of research software.

As the number of publications in this area continues to grow, it becomes increasingly difficult to organize, understand, and navigate the knowledge produced by the research community. To address this challenge, we have developed an AI4RSE taxonomy that captures concepts extracted from a large collection of scientific publications.

The taxonomy was generated using automated extraction techniques, Large Language Models (LLMs), and the IEEE Taxonomy (2025 edition) as a reference structure. The resulting taxonomy contains approximately 11,000 concepts organized into a hierarchical structure consisting of:

- High-Level Categories
- Middle-Level Categories
- Low-Level Categories
- Concepts

For example:

Artificial Intelligence

→ Computational Intelligence

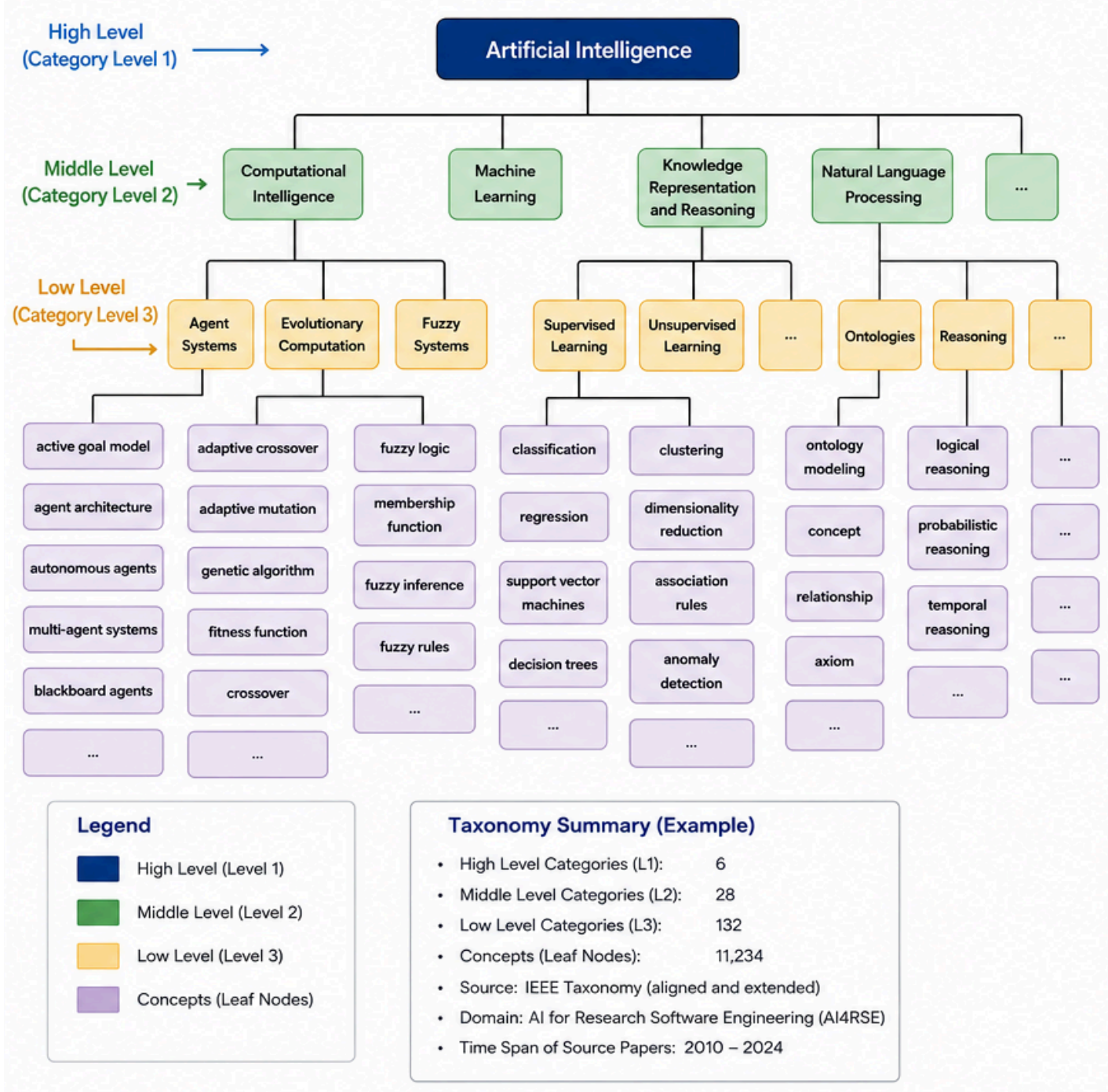
→ Agent Systems

→ Autonomous Agents

The initial categorization has already been completed. Every concept has been assigned to a category path within the taxonomy.

However, because the taxonomy was generated automatically, we cannot assume that all concept-category assignments are correct. Some concepts may have been assigned to inappropriate categories, some concepts may appear multiple times, and some concepts may not be relevant to the AI4RSE domain.

Before the taxonomy can be transformed into an ontology, we must first evaluate and refine the current categorization.



## 2. What is a Taxonomy?

A taxonomy is a structured classification system that organizes concepts into categories and subcategories.

The primary purpose of a taxonomy is to organize knowledge and group related concepts together. A taxonomy helps us understand how concepts are organized within a domain and provides a common vocabulary for discussing that domain.

A taxonomy focuses on classification and hierarchy. It answers questions such as:

- Which concepts belong together?
- Which category best represents a concept?
- How are concepts organized within a domain?
- Which concepts belong to a particular research area?

Unlike an ontology, a taxonomy primarily represents hierarchical relationships and classification structures. It does not explicitly represent richer semantic relationships, such as:

- supports
- improves
- depends on
- uses
- generates
- automates

In this project, the taxonomy serves as the foundation for ontology construction. Consequently, the quality of the ontology will depend heavily on the quality of the taxonomy.

## 3. Why Do We Need to Evaluate the Taxonomy?

Although the taxonomy has already been constructed, its quality has not yet been systematically evaluated.

Several types of issues may exist.

A concept may have been assigned to the wrong category.

For example, a concept related to Software Testing may have been assigned to Software Maintenance.

- A concept may appear multiple times under different categories.
- A concept may be overly generic and provide little value within the taxonomy.
- A concept may be ambiguous or poorly defined.

Some concepts may be valid Artificial Intelligence concepts, but may not be relevant to the AI4RSE domain.

If such issues remain in the taxonomy, they will also propagate into the ontology that we build in the next stage of the project. As a result, errors introduced during taxonomy construction may affect ontology quality, ontology evaluation, and any downstream applications that rely on the ontology.

For this reason, taxonomy evaluation is a critical first step before ontology construction.

## 4. Objective of This Task

The objective of this task is to evaluate the alignment between concepts and categories within the AI4RSE taxonomy and improve the quality of the current categorization.

We are not creating a new taxonomy.

We are not redesigning the IEEE Taxonomy.

We are not restructuring the overall hierarchy.

Instead, we are evaluating whether the existing concepts have been assigned to appropriate categories and identifying concepts that require modification. The outcome of this task should be a refined taxonomy that more accurately represents the AI4RSE domain and can serve as the foundation for ontology construction.

## 5. Taxonomy Evaluation Process

The taxonomy contains approximately 11,000 concepts. Reviewing every concept manually would be time-consuming and difficult to perform consistently.

Therefore, we will use Generative AI as an assistant during the evaluation process.

For each concept, we will provide:

- Concept name
- Concept definition
- Current taxonomy path

The taxonomy path consists of:

High-Level Category → Middle-Level Category → Low-Level Category

For example:

Artificial Intelligence → Computational Intelligence → Agent Systems

Generative AI will evaluate whether the concept belongs to the assigned category and whether a more appropriate taxonomy path exists.

The purpose of using AI is not to automate decision-making completely.

Instead, AI is used to:

- identify potentially problematic concepts
- suggest alternative taxonomy paths
- identify duplicate concepts
- identify ambiguous concepts
- provide an independent assessment of the current categorization

The final decision should always be based on our own judgment.

The goal is to identify concepts that require changes and record those changes in a structured format.

## 6. Types of Taxonomy Issues

During the evaluation process, we should pay particular attention to the following types of issues.

### Misclassified Concepts

A concept may have been assigned to an inappropriate category.

For example, a concept related to software testing may have been placed under software maintenance.

### Duplicate Concepts

The same concept may appear multiple times under different categories or under slightly different names.

### Ambiguous Concepts

Some concepts may have unclear meanings or may fit multiple categories equally well.

### Overly Generic Concepts

Some concepts may be too broad to be useful and may require further specialization.

### AI4RSE Relevance Issues

Some concepts may be valid Artificial Intelligence concepts, but may not contribute meaningfully to the AI4RSE domain.

Identifying these concepts will help improve the focus and usefulness of the taxonomy.

## 7. Using Generative AI for Taxonomy Alignment

Generative AI should be used to provide an independent assessment of concept-category alignment.

A suggested prompt is:

You are an expert in Artificial Intelligence, Software Engineering, Ontology Engineering, Knowledge Organization, and Taxonomy Engineering.

Context:

```
We are evaluating a taxonomy for Artificial Intelligence for Research Software Engineering (AI4RSE).
The current taxonomy is aligned with the IEEE Taxonomy (2025 edition). The categorization below was generated automatically and
now requires validation.
Your task is not to redesign the IEEE taxonomy. Instead, evaluate whether the concept is appropriately aligned with the current
taxonomy path.
Concept:
[Concept Name]
Definition:
[Concept Definition]
Current Taxonomy Path:
[High-Level Category → Middle-Level Category → Low-Level Category]
Please answer the following questions:
1. Does the concept belong to the assigned category path?
2. Is the categorization consistent with the IEEE Taxonomy?
3. If not, suggest a more appropriate category path.
4. Explain your reasoning in one short sentence.
5. Rate your confidence from 1 (very low) to 5 (very high).
Return your answer in JSON format:
{
  "alignment": "Correct | Partially Correct | Incorrect",
  "suggested_path": "...",
  "confidence": 1-5,
  "reasoning": "..."
}
```

The generated response should be treated as a recommendation rather than a final decision.

## 8. Recording Decisions in the Google Sheet

The Google Sheet will be the primary artifact produced during this task.

The purpose of the Google Sheet is to record all taxonomy-review decisions in a structured format that can later be processed automatically.

To keep the review process simple and scalable, we should only record three fields for concepts that require attention.

## Decision

Possible values are:

- Keep: Keep means the concept is correctly aligned and should remain unchanged.
- Move: Move means the concept should be reassigned to another taxonomy path.
- Merge: Merge means the concept is duplicated and should be merged with another concept.
- Remove: Remove means the concept should be excluded from the taxonomy.
- Discuss: Discuss means the concept requires additional review and cannot be resolved immediately.

## Reason

The reason should be short and concise.

Examples include:

- Fits current category.
- Better fits software testing.
- Duplicate concept.
- Too generic.
- Not relevant to AI4RSE.
- Requires expert review.

The purpose of this field is to provide traceability while keeping the review process efficient.

## New Taxonomy Path

This field should only be completed when the decision is Move.

Example: Artificial Intelligence → Machine Learning → Supervised Learning

For all other decisions, this field can remain empty.

## 9. Analysis

After the evaluation process is completed, we should analyze the collected data.

The analysis should answer questions such as:

- How many concepts were reviewed?
- How many concepts were flagged?
- How many concepts were moved?
- How many duplicates were identified?
- How many concepts were removed?
- Which categories contained the largest number of issues?
- Which issue types occurred most frequently?

The analysis should also discuss the usefulness and limitations of Generative AI for taxonomy alignment.

## 10. Deliverables

At the end of this task, we expect to produce:

1. A refined version of the AI4RSE taxonomy.
2. A Google Sheet containing all taxonomy-review decisions.
3. A summary of identified taxonomy issues.
4. Statistics describing the review process.
5. A specification for the automated taxonomy-refinement pipeline.
6. Recommendations for ontology construction in Task 2.

## 11. Expected Outcome

At the completion of this task, we expect to have a validated and refined AI4RSE taxonomy with improved concept-category alignment and reduced inconsistencies.

The resulting taxonomy will provide a more accurate representation of the AI4RSE domain and will serve as the primary input for Task 2, where the taxonomy will be transformed into an ontology by introducing semantic relationships, properties, constraints, and formal knowledge structures.

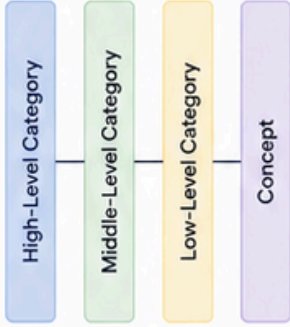
Estimated duration: 2 weeks.



# AI4RSE Taxonomy Alignment Evaluation and Refinement Process (Task 1)

## 1 Select a Concept for Review

Pick a concept from the taxonomy to evaluate.



## 2 AI-Assisted Evaluation

Provide the concept information and current taxonomy path to the AI assistant.



Inputs to AI:

- Concept Name
- Definition
- Current Taxonomy Path (High → Middle → Low)

AI returns a recommendation (JSON format) with alignment assessment and suggested path.

## 3 Review AI Recommendation and Make Decision

We review the AI suggestion and decide the appropriate action for this concept.

Possible Decisions:

- Keep (no change)
- Move (change category)
- Merge (duplicate concept)
- Remove (not relevant)
- Discuss (needs review)

## 4 Record in Google Sheet

Record ONLY the following three items for this concept.

- Decision**  
(Keep / Move / Merge / Remove / Discuss)
- Reason** (short)  
Why we made this decision.
- New Path** (only if Move)  
Provide the new taxonomy path (High → Middle → Low).

All records are stored in the shared Google Sheet.

## 5 Repeat for All Concepts (or in batches).

Repeat steps 1–4 for all concepts (or in batches).



## 6 Easy to Process and Revise

The structured decisions in the sheet allow automatic processing and easy revisions.

- Move concepts around
- Merge duplicates
- Remove irrelevant items

Re-run pipeline anytime to update the taxonomy.

## Input: Existing Taxonomy

AI4RSE Taxonomy (Aligned with IEEE Taxonomy 2025)  
~11,000 Concepts



## Google Sheet Structure (What We Record)

For each reviewed concept, we fill ONLY these three columns.

| Concept Name             | Current Path<br>(High → Middle → Low)       | WHAT WE RECORD         |                                      |                                                    |
|--------------------------|---------------------------------------------|------------------------|--------------------------------------|----------------------------------------------------|
|                          |                                             | Decision<br>(Required) | Reason<br>(Short)                    | New Path<br>(Only if Move)                         |
| Autonomous Agents        | AI → Comp. Intelligence → Agent Systems     | Keep                   | Fits current category.               | -                                                  |
| Automated Program Repair | AI → Machine Learning → Deep Learning       | Move                   | Better fits program repair.          | AI → Software Eng. → Program Repair                |
| Genetic Algorithm        | AI → Machine Learning → Supervised Learning | Move                   | Not supervised learning.             | AI → Comp. Intelligence → Evolutionary Computation |
| Autonomous Agents        | AI → Comp. Intelligence → Agent Systems     | Merge                  | Duplicate of Autonomous Agents.      | (Merge with ID: 12345)                             |
| Data Mining (General)    | AI → Data Engineering → Data Management     | Remove                 | Too generic, not specific to AI4RSE. | -                                                  |
| Cognitive Architecture   | AI → Cognitive Systems → Cognitive Modeling | Discuss                | Unclear scope, needs discussion.     | -                                                  |

Only rows with decisions other than "Keep" need to be recorded (Keep can be left blank to reduce noise).

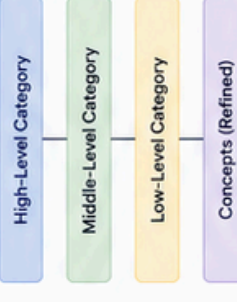
## Automated Processing Pipeline

The pipeline reads the Google Sheet and applies changes.



## Output: Refined Taxonomy

A cleaner, better-aligned AI4RSE taxonomy ready for Task 2 (Ontology Construction).



- Misclassified concepts moved
- Duplicates merged
- Irrelevant concepts removed
- Ambiguities reduced
- Higher quality and consistency

## Key Principles

✓ We make simple and consistent decisions.

✓ We record only 3 items (Decision, Reason, New Path).

✓ Everything is traceable in the Google Sheet.

✓ Easy to process, revise, and re-run anytime.

This refined taxonomy becomes the foundation for ontology construction.